Capital Circuit - Monopoly Game

CSC301 Group 4 | console Monopoly in Java (Abu Dhabi edition)

Rulebook and data-structure guide - demo revision

Purpose	Group revision for the demo (from 9 November 2026): every rule exactly as our game plays it, pointer-level drawings of every data structure, and model answers for the Section 8.2 questions
Our twists	T1 Reverse, T3 Expansion, T7 Recall - highlighted in green throughout this rulebook
Source	code*.java, code\data\board.txt and cards.txt as of 6 November 2026. Every rule was checked against the code; every scenario log is the exact output of the 96 automated tests
Run it	Eclipse: Run menu > Monopoly - Play / Play with seed / Twist demo / Tests (README.txt)
Brief reference	Section 3 (rules), 4-5 (structures, recursion), 6 and Appendix A (twists), 8 (demo)

Our three twists: T1 Reverse (turn queue reversed by a card) - T3 Expansion (three new squares at the end of turn 20) - T7 Recall (last three cards of a deck come to the top). Wherever a rule, square, card or drawing belongs to one of these twists, it is marked in green.

Part 1 - How to play Capital Circuit

1.1 The game at a glance

Item	Our game
Players	2 to 6, in the order their names are entered (join order)
Starting cash	1,500 AED each
Dice	Two six-sided dice; move forward by the total
Salary	200 AED every time you move forward onto or past GO (never when moving backward, never when sent to Jail)
Board	24 squares, clockwise from GO; 27 squares from the end of turn 20 (T3)
Turn limit	60 turns (one turn = one player's go, including any doubles)
Winner	The last player left, or the highest net worth when the turn limit is reached (net worth = cash + purchase price of every square owned)

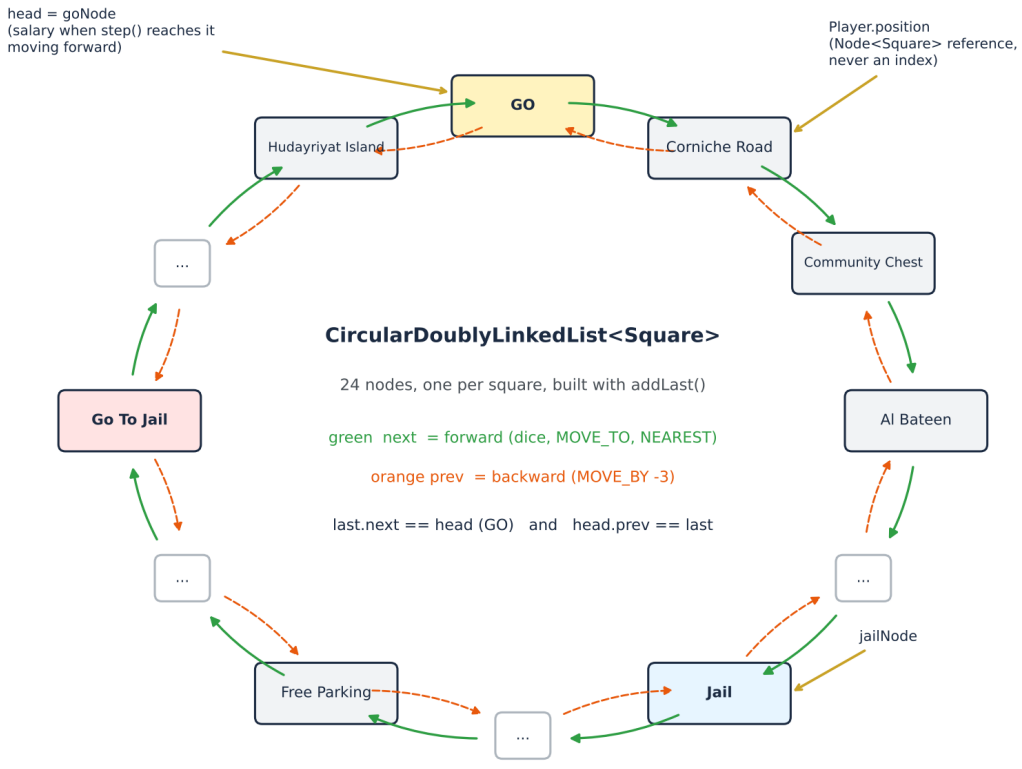
1.2 Twists - which ones are ours

The brief's pool has eight twists; each group implements three. Ours are the green rows. The other five are not in our game.

Twist	In our game?	What it does	When it happens	Log line
T1 Reverse	**YES**	The waiting players' order is reversed, so play runs the other way. The player who drew the card joins the back as usual.	A REVERSE card is drawn (C11 Chance, K11 Community)	REVERSE|player, then ORDER|p1|p2|...
T2 Earthquake	no	An EARTHQUAKE square destroys the next property	-	-
T3 Expansion	**YES**	Three new squares are linked into the board: Al Mina, Abu Dhabi Airport Express, Lulu Island. Nobody moves.	Automatically at the end of turn 20	EXPAND|new square|after square (x3)
T4 Rent chain	no	Rent chained along a colour group	-	-
T5 Mirage	no	A MIRAGE square moves the player back	-	-
T6 Inheritance	no	A bankrupt player's squares go to the other players	-	-
T7 Recall	**YES**	The last 3 cards of that deck move to the top, in order; RECALL goes to the bottom. With 3 or fewer other cards, nothing changes.	A RECALL card is drawn (C12 Chance, K12 Community)	RECALL|deck|id1|id2|id3
T8 Portals	no	Two PORTAL squares connected to each other	-	-

Note: the twist list for each group is set on Blackboard; check that ours is T1, T3, T7 before the demo.

1.3 The board



The 24-square starting ring, clockwise from GO. Green rows in the table: the three squares T3 adds at the end of turn 20.

Square	Type	Colour	Price	Rent (full group)	Note
GO	GO	-	-	-	Collect 200 when you move forward onto or past it
Corniche Road	PROPERTY	BLUE	60	4 (8)	
Community Chest	COMMUNITY	-	-	-	Draw a Community card

Square	Type	Colour	Price	Rent (full group)	Note
Al Bateen	PROPERTY	BLUE	60	6 (12)	
Al Mina (T3)	PROPERTY	BLUE	80	6 (12)	Added after Al Bateen at the end of turn 20: BLUE becomes 3 squares
Municipality Fee	TAX	-	-	pay 100	
Metro Red Line	TRANSIT	-	200	25+	
Jail	JAIL	-	-	-	Just visiting, or held in Jail
Yas Island	PROPERTY	GREEN	100	8 (16)	
Saadiyat Beach	PROPERTY	GREEN	120	10 (20)	
Chance	CHANCE	-	-	-	Draw a Chance card
Al Reem Island	PROPERTY	GREEN	140	10 (20)	
Metro Green Line	TRANSIT	-	200	25+	
Abu Dhabi Airport Express (T3)	TRANSIT	-	200	25+	Added after Metro Green Line: 4th transit
Free Parking	FREE	-	-	-	Nothing happens
Khalifa City	PROPERTY	ORANGE	160	12 (24)	
Community Fund	COMMUNITY	-	-	-	Draw a Community card
Masdar City	PROPERTY	ORANGE	180	14 (28)	
Etihad Rail	TRANSIT	-	200	25+	
Mussafah	PROPERTY	ORANGE	200	16 (32)	
Go To Jail	GO_TO_JAIL	-	-	-	Go straight to Jail (no salary)
Al Maryah Island	PROPERTY	RED	220	18 (36)	
Chance Desk	CHANCE	-	-	-	Draw a Chance card
Darb Toll Gate	TAX	-	-	pay 75	
Emirates Palace	PROPERTY	RED	260	22 (44)	
Hudayriyat Island	PROPERTY	RED	280	24 (48)	Last square of the starting board
Lulu Island (T3)	PROPERTY	RED	300	26 (52)	Added after Hudayriyat Island: new last square, its next is GO

Rent in brackets applies when the owner holds every PROPERTY of that colour. Transit rent: see 1.5.

1.4 Your turn, step by step

- **1 Your turn starts.** You are taken from the front of the turn queue.
- **2 In Jail?** Jail rules first (1.8); they decide whether and how you move.
- **3 Roll** two dice and move forward by the total. Passing or landing on GO pays 200.
- **4 Resolve the square** you land on (1.5). A card that moves you means the new square is resolved too.
- **5 Rolled a double?** Roll again (back to 3). A third double in the same turn: you do not move and go to Jail.
- **6 Sent to Jail** at any point? Your turn ends at once, even after a double.
- **7 Turn ends.** If you are not bankrupt, you join the back of the queue. **T3:** if this was turn 20, the three new squares are added now.
- **8 Game over?** It ends when only one player is left, or after turn 60.

1.5 Squares, buying and rent

You land on	What happens
An unowned PROPERTY or TRANSIT	You may buy it for its price if you can afford it (you are asked Y/N). If you do not buy it, it stays with the bank - no auctions.
A square someone else owns	Pay the rent to the owner - even if the owner is in Jail.
Your own square	Nothing.
PROPERTY rent	The base rent, doubled if the owner holds every PROPERTY of that colour. After T3, BLUE has 3 squares (Al Mina) and RED has 4 (Lulu Island), so a full set needs the new square too.
TRANSIT rent	25 x 2^(transit squares the owner holds - 1): 1 = 25, 2 = 50, 3 = 100, 4 = 200 (the 4th exists only after T3).
TAX	Pay the amount to the bank: Municipality Fee 100, Darb Toll Gate 75.
CHANCE / COMMUNITY	Take the top card of that deck, do what it says, put it at the bottom (1.6).
GO TO JAIL	Go straight to Jail; no salary; your turn ends.
GO, JAIL (visiting), FREE PARKING	Nothing.

No houses, hotels, mortgages, trading or auctions (brief 3.4).

1.6 The cards

Each deck is shuffled once at the start (the seed decides the order). Drawing takes the top card; after it is used it goes to the bottom. A Get out of Jail free card stays with you until you use it, then goes to the bottom of its deck.

Chance (12 cards)

Card	Text	Effect
C1	Advance to GO	Move forward to GO (salary 200)
C2	Go back 3 squares	Move back 3; no salary
C3	Take the nearest metro	Move forward to the next TRANSIT square
C4	Speeding fine on Sheikh Zayed Road	Pay 15 to the bank
C5	Visit the Louvre: advance to Saadiyat Beach	Move forward to Saadiyat Beach
C6	Grand Prix weekend: advance to Yas Island	Move forward to Yas Island
C7	Bank dividend paid to you	Collect 50
C8	Get out of Jail free	Keep it until you are in Jail
C9	Advance to Hudayriyat Island	Move forward to Hudayriyat Island
C10	Move forward 2 squares	Move forward 2
C11 (T1)	Road works on the E11: turn order reversed	REVERSE: the waiting players' order is reversed
C12 (T7)	Old files reopened: last 3 Chance cards to the top	RECALL: last 3 Chance cards move to the top

Community (12 cards)

Card	Text	Effect
K1	Research grant awarded	Collect 200
K2	Host a Corniche picnic for everyone	Pay 50 to each other player
K3	Birthday gifts from each player	Collect 10 from each other player
K4	Go directly to Jail	Go to Jail (no salary)
K5	Get out of Jail free	Keep it until you are in Jail
K6	Hospital fees	Pay 100 to the bank
K7	Utility refund from the municipality	Collect 20
K8	Advance to GO	Move forward to GO (salary 200)
K9	Evening walk: advance to Corniche Road	Move forward to Corniche Road - from a Community square this is almost a full lap, so you pass GO (salary)
K10	School fees	Pay 50 to the bank
K11 (T1)	Majlis etiquette: elders first - turn order reversed	REVERSE

Card	Text	Effect
K12 (T7)	Lost and found: last 3 Community cards to the top	RECALL

"Move forward to X" cards always go forward, so passing GO on the way pays the salary. Pay-each / collect-each go round the other players in the current turn-queue order (after a REVERSE, the reversed order).

1.7 Doubles

- A double (both dice the same) gives another roll after the square is resolved.
- Third double in the same turn: you do **not** move and go straight to Jail.
- If a double lands you in Jail (Go To Jail square or card), the turn ends - no extra roll.

1.8 Jail

- You go to Jail from the GO TO JAIL square, a Go to Jail card, or a third double. You are placed on Jail directly: no salary.
- At the start of each turn in Jail: **1** if you hold a Get out of Jail free card, it is used automatically - you leave and take a normal turn; **2** otherwise roll: a double frees you and you move by that roll (no extra roll for that double); **3** no double: you stay. On the **third** failed try you pay 50 and move by that roll.
- You still collect rent while in Jail.

1.9 Paying when you are short - bankruptcy

- If you must pay more than you have, you pay **everything you have** and are **bankrupt** - no selling, no borrowing.
- Your cash goes to whoever you owed: the square's owner for rent, the bank for tax, fines and Pay cards.
- All your squares go back to the bank, unowned. Your Get out of Jail free cards go to the bottom of their decks.
- You leave the game and are removed from the turn queue; the others keep their order.
- Pay 50 to each player (K2): you pay one player at a time, in queue order. If you cannot pay someone in full, they get what you have left and you are bankrupt; later players get nothing.
- Collect 10 from each player (K3): a player who cannot pay you in full pays what they have and is bankrupt to you; the others still pay.

1.10 End of the game and the ranking

- Checked after every turn: **one player left** (LAST_PLAYER) or **turn 60 reached** (MAX_TURNS).
- Ranking: players still in the game by net worth (cash + purchase price of every square they own), highest first; a tie goes to the player who joined first (even after a REVERSE). Bankrupt players follow, the most recent bankruptcy first.
- Pressing Q stops the game early and shows the standings; no END line is written to the log.

1.11 Playing in Eclipse

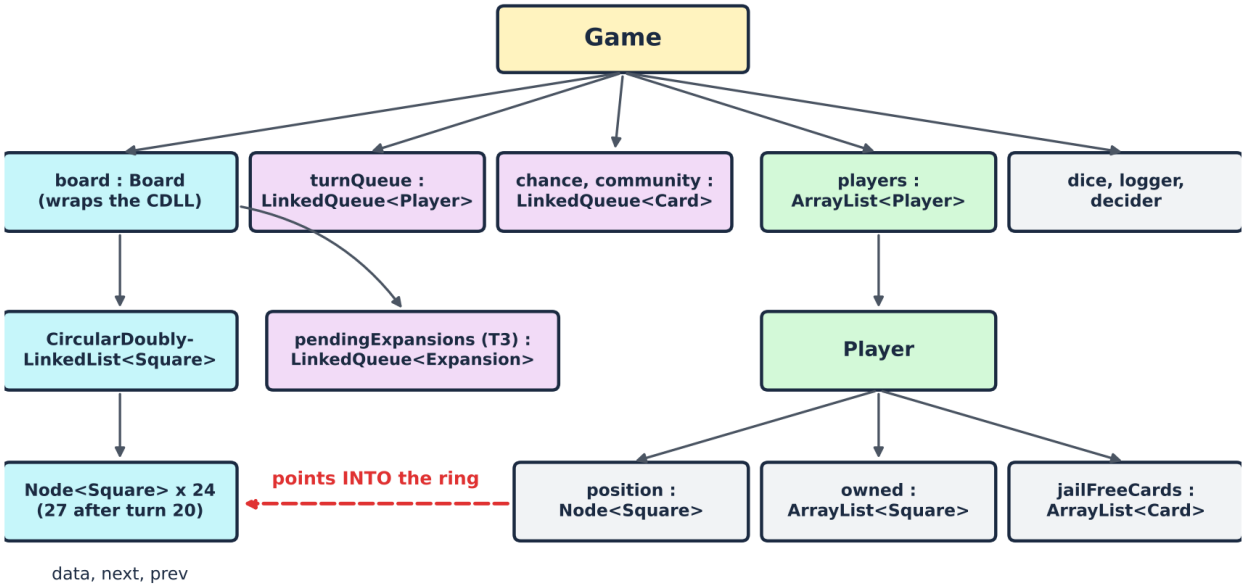
Run menu entry	What it does	Controls
Monopoly - Play	Asks for player names, then shows the colour board each turn	Enter = roll, Y/N = buy, Q = quit; click in the Console first
Monopoly - Play with seed	As Play, plus a seed: the same seed gives the same dice and cards	As above
Monopoly - Twist demo	Seed 73, 4 players, plain log lines: RECALL at turn 3, REVERSE at turn 8, EXPAND after turn 20	Runs by itself
Monopoly - Tests	96 tests, then the constraint check	Expect 96 passed, CONSTRAINT CHECK: PASS

Part 2 - The data structures, drawn

The brief fixes which structure does which job: the **board** is our own circular doubly linked list, **turn order** and the **two decks** are our own linked queue, **ArrayList** holds indexed data that changes size, and **recursion** (no loops) does movement, doubles, landing, rent, net worth and searches. Draw these from memory before the demo - the instructor asks for links before and after an operation.

2.1 The whole picture

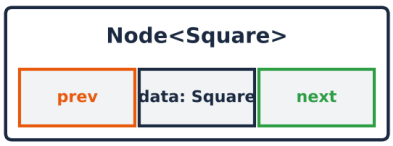
Who holds what - the objects of one running game



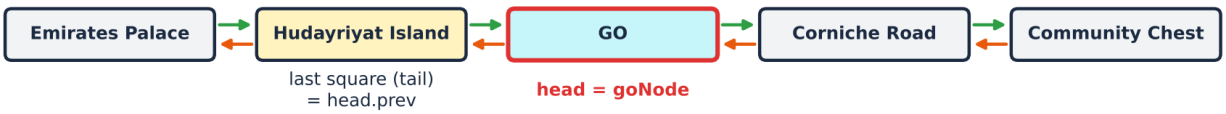
Player objects are shared: the same object is in players, in turnQueue (unless playing or bankrupt) and is the owner of its squares.

2.2 Circular doubly linked list - the board

Node<Square> and the links around GO (exam favourite)



One node per square. A player's position is a reference to one node, never an index. Only Game.movePlayer and Game.sendToJail change it.



Hudayriyat Island.next == GO **GO.prev == Hudayriyat Island** **(there is no null anywhere in the ring)**

next **prev**

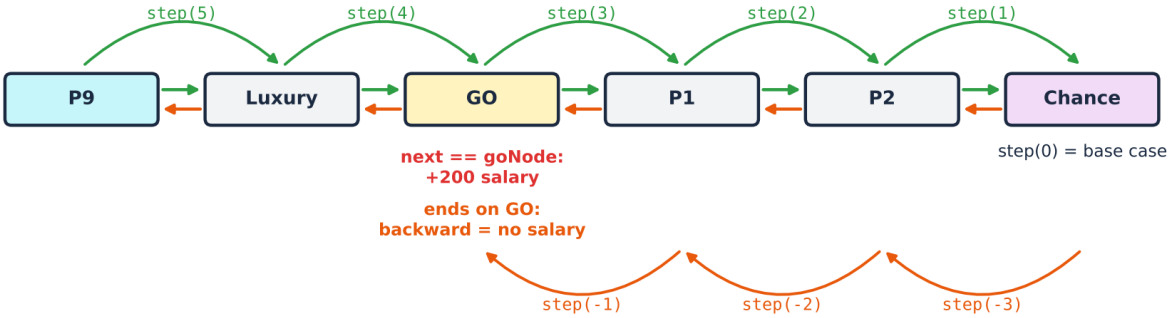
After T3 (turn 20): Lulu Island is inserted after Hudayriyat Island, so Lulu Island.next == GO and GO.prev == Lulu Island.

Question	Answer
Why not an ArrayList?	Movement would need index arithmetic (% size, forbidden for movement) and inserting a square (T3) would shift every later index - O(n). With nodes, a move is one hop and an insert changes 4 links.
Why doubly linked?	Backward moves (Go back 3) follow prev in O(1) per step; a singly linked list would need almost a full lap.
Why circular?	There is no end: next of the last square is GO, so moving round needs no special case.

Question	Answer
Cost	addLast, insertAfter, remove, getHead, size: O(1). Moving k squares: O(k). Searching by name or type: O(n).

step(): forward over GO (salary), then a card moves back 3 (no salary)

Brief 8.2 scenario on the test board: A on P9 (two squares before GO) rolls 2 + 3 = 5, lands on Chance, draws "Go back 3 squares".




```
Log:  TURN|1|A   ROLL|A|2|3   MOVE|A|P9|Chance   SALARY|A|1700   CARD|A|C2   MOVE|A|Chance|GO
```

Answer: one salary (passing GO forward). Landing on GO backwards pays nothing; GO then does nothing.

Base case: steps == 0 returns the node. steps > 0 follows next and counts down; steps < 0 follows prev and counts up.

T3 insertAfter - the twist's pointer change

T3 Expansion - CircularDoublyLinkedList<Square>.insertAfter at the end of turn 20

Before: ... Al Bateen <-> Municipality Fee ...



Player on Al Bateen holds a reference to that node only

After EXPAND|Al Mina|Al Bateen (4 links set, O(1); nobody moves)



n.prev = node; n.next = after
node.next = n; after.prev = n

next prev

Inserting after the LAST square: EXPAND|Lulu Island|Hudayriyat Island



dashed: the rest of the ring
GO -> Corniche Road -> ...
-> Hudayriyat Island

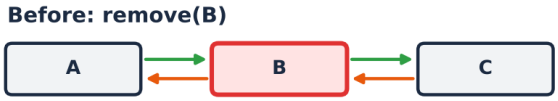
Lulu.next = GO and GO.prev = Lulu, so step() still pays the salary when it reaches goNode; size() = 27, so searches and the board view adapt.

Step	Link set (node = Al Bateen, after = Municipality Fee, n = Al Mina)
1	n.prev = node
2	n.next = after
3	node.next = n
4	after.prev = n; size++

Order matters: read after = node.next before changing node.next, or the old successor is lost.

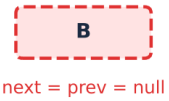
remove(node) - drawing the links before and after (pointer-tracing practice)

In the list since the core build and tested; not used in play (T2 Earthquake is not our twist), but a likely tracing question.





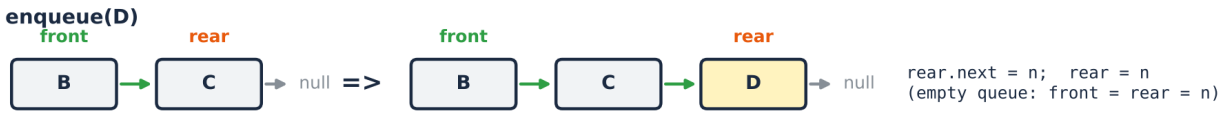

```
1  B.prev.next = B.next     (A.next = C)
2  B.next.prev = B.prev     (C.prev = A)
3  if B == head: head = B.next
4  B.next = B.prev = null;  size--
```



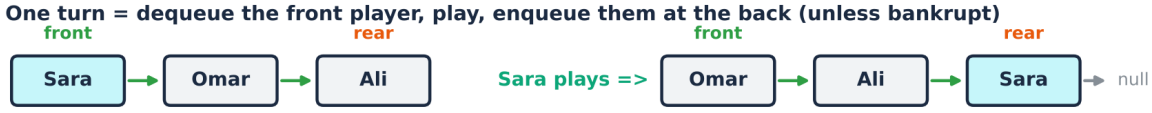
O(1): only the two neighbours change. A player standing on B would have to be moved first - in our game nothing calls remove().

2.3 Linked queue - turn order

LinkedQueue<T>: QNode (data, next) with front and rear





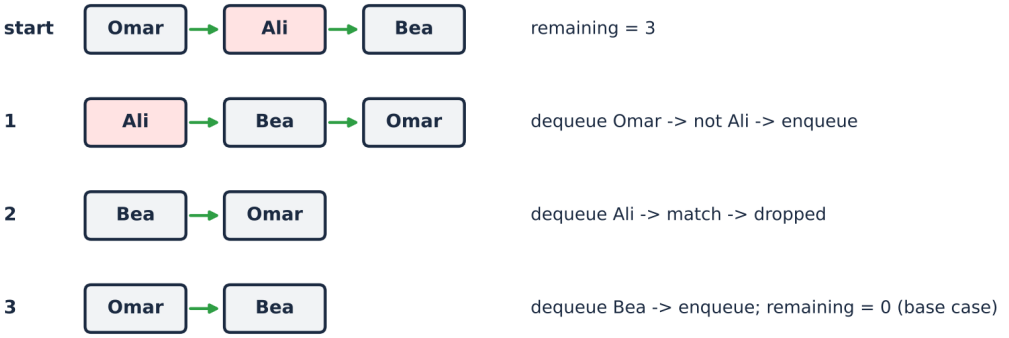


Empty queue: dequeue() and peek() throw IllegalStateException("... empty queue"). All operations O(1).

Why not ArrayList? remove(0) shifts every element: O(n). ArrayDeque / java.util.LinkedList are banned by the brief.

Removing a bankrupt player from the middle: one full rotation (remove -> removeRotate)

Queue Omar, Ali, Bea; Ali goes bankrupt. size = 3, so removeRotate runs 3 times: dequeue x; keep it (enqueue) unless it is Ali.

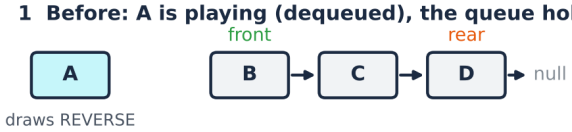


Result Omar, Bea: same relative order as before, Ali gone. O(n), uses only dequeue and enqueue - no index into the queue.

T1 reverse - the twist on the turn queue

T1 Reverse - LinkedQueue<Player> turnQueue (brief 8.2 scenario)

1 Before: A is playing (dequeued), the queue holds B, C, D



2 QueueTwists.reverse(q): each call dequeues one player; the call stack holds them



3 After: REVERSE|A, ORDER|D|C|B; A is enqueued at the back when the turn ends



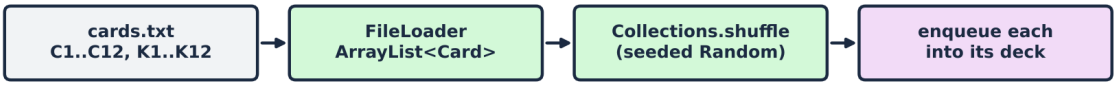
Next four turns: D, C, B, A

Cost: O(n) queue operations

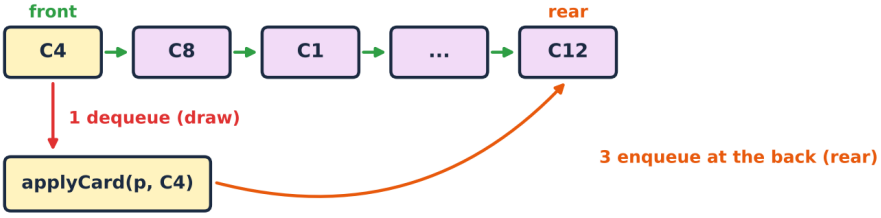
Why it obeys the T1 rule: reverse() calls only dequeue, enqueue and isEmpty. The players removed on the way down wait on the **call stack**, not in a list, array or stack object. Base case: empty queue. It stops because each call removes one player.

2.4 Linked queue - the card decks

One card's life: file -> ArrayList -> shuffle -> queue -> draw -> apply -> back of the deck



Chance deck while playing (front first):



2 apply the effect (move, pay, collect, jail, REVERSE, RECALL...)

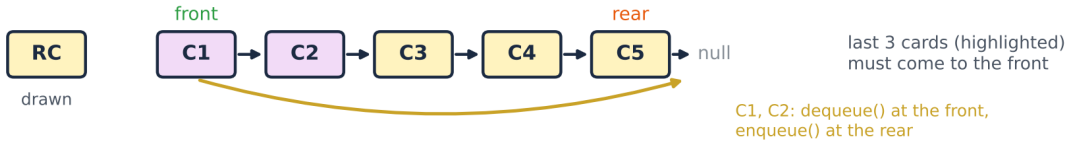
REVERSE (T1) and RECALL (T7) are applied in step 2 and then also go to the back.
If a card moves the player, landOn() runs again for the new square (depth at most 2).

JAIL_FREE: kept in the player's ArrayList<Card>; enqueued only when used or on bankruptcy

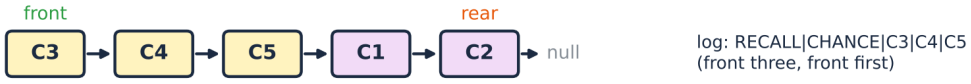
T7 rotate - the twist on a deck

T7 Recall - LinkedQueue<Card> (Chance deck), RECALL drawn with 5 other cards

1 RECALL (RC) dequeued and applied: n = deck.size() = 5 > 3



2 rotate(deck, n - 3) = rotate(deck, 2): two dequeue/enqueue steps, recursive



3 RECALL goes to the back as usual (normal card return)



Why n - 3: after RECALL is drawn the deck holds n cards. Moving the first n - 3 to the back leaves the last 3 at the front, still in order. rotate(q, k) moves one card per call and stops when k = 0.

2.5 ArrayList - indexed data that changes size

ArrayList	Holds	Why an ArrayList
Player.owned	The squares a player bought	Indexed recursion (netWorth(owned, index)); add at the end O(1); returning squares on bankruptcy removes from the end O(1)
Game.players	All players in join order	Sorted for the ranking with Collections.sort (Player implements Comparable)
Player.jailFreeCards	Held Get out of Jail free cards	Small list that grows and shrinks
Card list (loading)	All cards in file order	Collections.shuffle needs a list; then enqueued into the decks

Why not a linked list for owned squares? We need get(index) for the recursive net worth and sorting; an ArrayList gives O(1) get, and java.util.LinkedList is banned anyway.

2.6 Recursion - every method that must be recursive

Recursion traces - the calls, the base case, and why each one stops

takeTurn - two doubles, then a non-double (brief 8.2)

```
takeTurn(A, 0)          rolls 1+1 double   -> move, landOn
  takeTurn(A, 1)        rolls 2+2 double   -> move, landOn
    takeTurn(A, 2)      rolls 1+2          -> move, landOn; not a double: STOP
```

A third double at doublesCount == 2 goes to Jail without moving. doublesCount rises each call, so at most 3 calls.

landOn - a card that moves the player

```
landOn(A)               Chance -> C2 "Go back 3": movedByCard = true
  landOn(A)             GO: nothing happens, movedByCard = false: STOP
```

No card move ends on a CHANCE/COMMUNITY square (tested before and after T3), so the depth is at most 2.

transitRent(3) - owner has 3 transit squares

```
transitRent(3)          = 2 x transitRent(2) = 100
  transitRent(2)        = 2 x transitRent(1) = 50
    transitRent(1)      owned <= 1: return 25   (base case)
```

25 x 2^(n-1): 25, 50, 100, and 200 with the 4th transit added by T3.

netWorth(owned, 0) - Sara owns Yas Island (100) and Mussafah (200)

```
netWorth(owned, 0)      100 + netWorth(owned, 1) = 300
  netWorth(owned, 1)    200 + netWorth(owned, 2) = 200
    netWorth(owned, 2)  index >= size: return 0   (base case)
```

Total net worth = cash + 300. The index rises to the list size, so it stops.

Method	Base case	Recursive case	Why it stops
step (Board)	steps == 0	hop next (steps - 1) or prev (steps + 1)	|steps| shrinks by 1
takeTurn (Game)	3rd double, jailed/bankrupt, or not a double	takeTurn(p, doublesCount + 1)	at most 3 calls
landOn (Game)	the square did not move the player	landOn(p) after a card move	card moves never end on a card square: depth <= 2
transitRent	owned <= 1: 25	2 x transitRent(owned - 1)	owned shrinks to 1
netWorth (Player)	index >= size: 0	price + netWorth(owned, index + 1)	index grows to size
ownsGroup (Board)	missing square of the colour: false; back at start: true	check next node	the ring is finite
findByName / findNearest	remaining == 0: null; match: node	next node, remaining - 1	remaining shrinks
reverse (T1)	queue empty	dequeue x, reverse(rest), enqueue x	one fewer element each call
rotate (T7)	k <= 0 or empty	enqueue(dequeue()), rotate(q, k - 1)	k shrinks by 1
applyExpansions (T3)	no pending EXPAND line	dequeue one, insertAfter, recurse	3 lines, one per call

Part 3 - Demo revision

3.1 The brief's scenarios, answered with our real log output

These are the Section 8.2 scenarios. The logs are the exact output checked by tests\GameTests.java (they run on tests\data\test-board.txt, a 20-square board with squares P1-P9, Chance, Tax, Jail ...).

"A player two squares before GO rolls 5, lands on Chance: Go back 3 squares."

```
Log

TURN|1|A
ROLL|A|2|3
MOVE|A|P9|Chance
SALARY|A|1700
CARD|A|C2
MOVE|A|Chance|GO
```

One salary - for passing GO forwards. The backward move lands on GO but pays nothing; GO does nothing. See drawing 2.2 (step).

"Sara owes 300 rent, has 120 AED and a JAIL_FREE card."

```
Log

TURN|1|Sara
ROLL|Sara|1|3
MOVE|Sara|P2|P3
RENT|Sara|Omar|120|0|1620
BANKRUPT|Sara|Omar
```

Omar gets her 120 (the RENT line shows what was actually paid). Her squares go back to the bank, the JAIL_FREE card goes to the bottom of its deck, and she is not re-queued: queue Omar, Ali.

"Sara is playing, queue Omar then Ali. She draws COLLECT_EACH 10; Omar has 5 AED."

```
Log

CARD|Sara|K3
CASH|Omar|-5|0
CASH|Sara|+5|1505
BANKRUPT|Omar|Sara
CASH|Ali|-10|1490
CASH|Sara|+10|1515
```

Queue order: Omar first. He pays 5 and is bankrupt to Sara (removed by rotation, drawing 2.3); Ali still pays 10. Queue before: Omar, Ali (Sara playing). After the turn: Ali, Sara.

"Jailed, no JAIL_FREE, failed twice. Third attempt rolls 2 and 3 with 30 AED."

```
Log

TURN|1|A
JAIL_ROLL|A|2|3|FINE
CASH|A|-30|0
BANKRUPT|A|BANK
```

Third failure means the 50 fine. A can pay only 30, so pays 30 and is bankrupt to the bank; with two players the game ends: LAST_PLAYER.

"Trace takeTurn() for two doubles, then a non-double."

```
Log

TURN|1|A
ROLL|A|1|1
MOVE|A|GO|P2
ROLL|A|2|2
MOVE|A|P2|P3
ROLL|A|1|2
MOVE|A|P3|P4
```

takeTurn(A,0) -> takeTurn(A,1) -> takeTurn(A,2); the third roll is not a double, so the recursion stops. Drawing 2.6.

(T1) "A draws REVERSE while the queue holds B, C, D. Who plays the next four turns?"

```
Log

TURN|1|A
ROLL|A|1|2
MOVE|A|GO|Chance
CARD|A|R1
REVERSE|A
ORDER|D|C|B
TURN|2|D    TURN|3|C    TURN|4|B    TURN|5|A
```

D, C, B, A. reverse() turns B, C, D into D, C, B; A is not in the queue during the turn and is enqueued at the back when it ends.

3.2 Questions to expect - short model answers

Question	Model answer
Draw the links around GO.	Last square.next = GO and GO.prev = last square (Hudayriyat Island; Lulu Island after turn 20). No nulls in the ring.
Where is a player's position stored, and who may change it?	Player.position, a Node<Square>. Only Game.movePlayer (via Board.step) and Game.sendToJail change it.
How does the queue remove a bankrupt player from the middle?	remove() rotates the whole queue once: dequeue each player and enqueue it again, except the bankrupt one. Order kept, O(n).
Why ArrayList for a player's squares?	Indexed recursion (netWorth), sorting and adding at the end; O(1) get. A linked list gives no index access and java.util.LinkedList is banned.
Walk me through a turn, class by class.	Game.playOneTurn (dequeue) -> jailTurn or takeTurn -> Dice.roll -> movePlayer -> Board.step -> landOn -> Square.land (Property/Transit/Tax/Card/Corner) -> back in playOneTurn: enqueue, T3 at turn 20, end check. GameLogger writes each line.
Follow one Chance card from the file back to its deck.	cards.txt -> FileLoader (ArrayList) -> Collections.shuffle -> enqueue -> CardSquare.land -> Game.drawCard (dequeue) -> applyCard -> enqueue at the back (JAIL_FREE: held by the player first).
Can landOn() recurse forever?	No: it only recurses after a card move, and no card move ends on a CHANCE or COMMUNITY square - checked by a test on the 24-square and the 27-square board. Depth at most 2.
What does step() do with 0 or a negative number?	0 is the base case: it returns the current node. Negative follows prev and counts up to 0, paying no salary.
If I add a new square type, what changes?	A new Square subclass with land(), one case in FileLoader.parseSquare, and the board file. Game, Board and the list stay the same.
(T1) How does reverse() work without a list or stack?	Dequeue one, reverse the rest recursively, enqueue that one. The call stack holds the players; only enqueue, dequeue, isEmpty are used.
(T3) Two players stand on Al Bateen when Al Mina is inserted. What happens?	Nothing to them: they hold the Al Bateen node; its next is now Al Mina. Four links change, nobody moves, no salary.
(T3) What happens to Sara's double BLUE rent at turn 20?	Al Mina is a new unowned BLUE square, so ownsGroup now fails and her rent drops back to base until she buys Al Mina.
(T7) The Chance deck has RECALL plus 3 other cards. What happens?	Nothing changes and no RECALL line is printed; RECALL goes to the bottom as usual.
Which constraint did the AI ignore most often?	Using classes not on the allowed list (java.util.Arrays twice, AI log cases 4 and 6). We enforce it with ConstraintCheck, which runs after every test run.

3.3 Practise on paper (no peeking)

- Draw the four nodes Hudayriyat Island, Lulu Island, GO, Corniche Road after turn 20 with every next and prev arrow.
- Draw the turn queue (front, rear, every QNode.next) for Sara, Omar, Ali, Bea; then after Omar goes bankrupt; then after Ali draws REVERSE.
- Just after RECALL is drawn, the Chance deck is C5, C9, C2, C7, C1 (front first). Write the deck afterwards and the log line. (Answer: n = 5, rotate(deck, 2) gives C2, C7, C1, C5, C9, then RECALL at the back; log RECALL|CHANCE|C2|C7|C1.)
- Trace transitRent for an owner with 4 transit squares after T3 (answer: 200).
- Lands on Chance Desk, draws C1 Advance to GO: trace landOn, count salaries. (Answer: depth 2, one salary.)